IHear- Mobile Application for Hearing- Impaired Children to Facilitate Their Hearing and Speech Abilities

Koralage H.C
Department of Computer Systems Engineering
Sri Lanka Institute of Information Technology
Malabe, Sri Lanka
it20149758@my.sliit.lk

Liyanage G.N.J
Department of Computer Systems Engineering
Sri Lanka Institute of Information Technology
Malabe, Sri Lanka
it20075712@my.sliit.lk

Senanayake S.V.M
Department of Computer Systems Engineering
Sri Lanka Institute of Information Technology
Malabe, Sri Lanka
it20024604@my.sliit.lk

Mapa M.H.M.I.D
Department of Computer Systems Engineering
Sri Lanka Institute of Information Technology
Malabe, Sri Lanka
it20468460@my.sliit.lk

Buddhima Attanayaka
Department of Computer Systems Engineering
Sri Lanka Institute of Information Technology
Malabe, Sri Lanka
buddhima.a@sliit.lk

Hansika Mahaadikara
Department of Computer Systems Engineering
Sri Lanka Institute of Information Technology
Malabe, Sri Lanka
hansika.m@sliit.lk

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***Abstract*— Children with hearing impairments face serious challenges as they are unlikely to develop normal speech and language abilities. It is important to build positive relationships with the external environment to support hearing impaired children's learning; hence, IHear was introduced. 'IHear' is a mobile application meticulously developed that aims to support children with mild to moderate hearing impairments under the age group of 6–10. The IHear mobile application serves as a potent intervention designed to facilitate effective learning while mitigating potential academic setbacks. A positive relationship with the surrounding environment that incorporates learning is crucial for hearing-impaired children since the development aspects of a child are directly connected to external factors. This study investigates IHear's tailored and comprehensive approach to speech and learning development using technology and machine learning.**

Keywords—Hearing-impaired children, hearing and speech abilities, Educational software application, Assistive learning, speech and hearing impairments, machine learning, sound classification, sound amplification, personalized learning, real-time support

I. INTRODUCTION

Hearing impairment is defined as an individual's inability to perceive sounds. This comprises individuals with any degree of congenital or acquired hearing loss. The author of the article [1] states that the World Health Organization (WHO) estimates that 9 percent of Sri Lankans suffer from hearing loss and 15 percent are entirely deaf. According to recent statistics, the population with hearing impairments will increase to considerable levels, from 131 million to 267 million by 2050 in the South Asia region [2]. The journey of children with hearing and speech impairments is fraught with challenges that result in diminished quality of life. The communication problems, meshed with low academic achievements, stand to reduce the chances of getting any further education or building a career. It is important to detect hearing impairment early on. Interventions to enhance the quality of life for children with hearing impairments are now standard practice [3].

In order to promote social development, it is necessary to enhance linguistic proficiency. As distinct levels of hearing have been identified, the effects of the constant improvement vary accordingly. In noisy environments, hearing-impaired children encounter one of the greatest obstacles to speech perception. Even with the development of hearing aid technology, this remains one of the problems [4].

This study intends to provide an advanced learning environment for children in the age group of 6–10, focusing on building a positive relationship with their auditory and speech abilities. The IHear mobile application is developed with state-of-the-art technologies, including machine learning, to provide a comprehensive set of features that aim to improve communication capabilities. IHear bridges the gap between hearing and speech by allowing children to learn at their own pace. Allowing continuous screening, personalized sound amplification and transcript generation, sound classification, and speech accuracy checking activities to consistently check the improvements.

The IHear research paper goes into the technical aspects of the algorithms used and the effectiveness of identifying the sounds. The goal of the current study is to assess the child's hearing development and tailor learning activities to meet their unique needs. The application has a significant difference as it consistently engages the target audience by making the application more user friendly.

IHear has the potential to empower children with speech and hearing impairments, providing them with the necessary tools and resources to enhance their communicative skills and participate actively in academic and social domains. The purpose of this study is to shed light on the prospective benefits of this technology and to encourage additional research and development in the field of assistive technology. This research has the potential to have a significant effect on the field of special education by enhancing the lives of children with speech and hearing impairments.

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II. Related Work

The following related studies support the identification of the need for the IHear application for mild to moderate hearing-impaired children. The utilization of mobile health (m-health) approaches to enhance accessibility, scalability, and service quality in addressing hearing loss underscores the critical need for thorough implementation evaluations, particularly in lower- and middle-income countries [5]. This review provided a comprehensive understanding of the features, functions, and effectiveness of existing apps. Based on this, Johnson and Williams [6] designed and tested a mobile app for speech therapy in hearing-impaired children. They assessed the application's usability, engagement, and efficacy to determine its pros and drawbacks.

A study conducted by J. N. Polvanov examines the need to individualize the education of hearing-impaired children [7]. It was stated that it is essential to tailor learning methods and technologies to the needs of children. In the study conducted by Park et al. [8], a speech enhancement algorithm was proposed to improve speech quality in a hearing aid environment by applying noise reduction algorithms with deep neural network learning based on noise classification.

In [9], the research assesses the usability of an augmented reality application investigated augmented reality (AR) applications for speech therapy. The study demonstrates a high level of usability among speech therapy specialists.

The Hearing screening using an audiometry application for hearing impairment research paper by E. Kiktová et al emphasizes the significance of screening in a learning environment for children [10]. This research was extensively read and is the foundation for one of the IHear application's features. As the title of the paper suggests, our research paper also takes very specific auditory stimuli into consideration. This report highlights the value of routine hearing assessments by emphasizing their contribution to ongoing education and personal development. The study's findings highlight the significance of incorporating hearing screening into educational programs for children with hearing impairments and provide helpful guidance on how to implement it.

There are studies that discuss adjusting hearing aid parameters to assist children with this impairment and enable them to recognize environmental sounds [11]. It is essential for children to have auditory awareness in order to learn from their surroundings and receive balanced exposure to the outside world.

VOIS Mobile is a mobile application designed to aid in the speech therapy of hearing-impaired children in Myanmar [12]. It discusses the use of CNN (Convolutional Neural Network) by mobile applications to recognize children's speech and speech-related challenges. The experimental results are nearly 60% and state that simple practice of speech improves the quality of life of children with hearing loss in Myanmar.

III. Methodology

After a thorough investigation of which features are most suitable for an IHear application, four components were identified that are capable of increasing the scope according to the needs of the children. As the research focuses on children with mild to moderate hearing loss, these children have a hearing level that indicates a considerable level of awareness of their surroundings. However, their ability to perceive speech is poor. This research paper aims to evaluate the effectiveness and impact of the IHear learning application for children with speech and hearing impairments. To achieve the goal that was stated earlier, comprehensive techniques are utilized under the below-explained components.

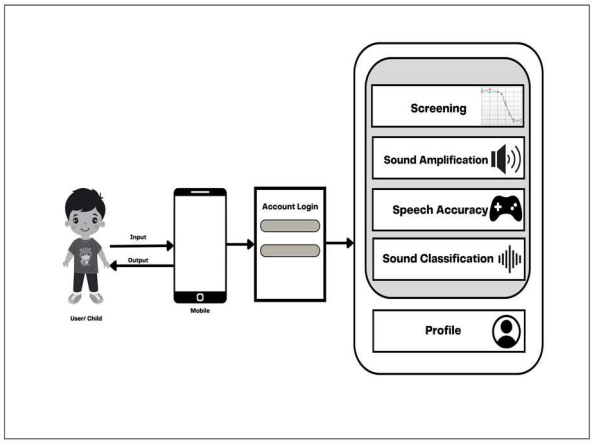


Fig. 1. Overall High-Level System Diagram

A. *Hearing Screening and ability screening for Impaired children to identify the level of hearing discrepancies and communication*

Screening of hearing-impaired Children is done in two components to add depth to the functionality.

1) Screening of Level of Hearing Impairment

The study focuses on the analysis of the Screening functionality, which has been divided into different levels. The aim of this research is to examine the current state of screening functionality and identify any potential issues or areas for improvement. The methodology employed in this study involves a thorough examination of the segments of the Screening functionality, with a particular emphasis on their functionality. This study aimed to assess the hearing ability of participants using a multi-faceted approach. Through the application, it plays tones with different frequencies to identify the hearing level of the user. The frequencies which are used for the test are 250Hz, 500Hz, 1000Hz, 2000Hz, and 4000Hz. There are certain levels of the test and from level to level the decibel range of the selected tones is increased and played through the application. The user has to identify all the tones they heard and through that the system analyzes the hearing stage of the user and provides output. Following the completion of these tasks, a graph is generated to display the participant's score. All data is then analyzed, and participants were classified according to their level of hearing impairment, ranging from normal to profound.

2) Screening of children's ability to communicate using sign language.

Real-time gesture recognition has been integrated into the created mobile application's interactive user interface to aid with sign language learning. When the function is first launched, users are shown a word on the screen along with an illustration of a sign language gesture that graphically represents the word. Users are asked to replicate the gesture by making it in front of the camera of their device. The application records the user's gesture, analyses it using an

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adapted version of the Xception convolutional neural network (CNN) model, and then forecasts the matching sign language gesture. The expected gesture corresponding to the displayed word is then contrasted with this prediction.

In addition, the application rates each user's performance numerically, indicating how accurately they classified gestures. The application's functionality and design are focused on giving users an engaging, instructive, and interactive experience that aids them in their quest to learn sign language.

Images of sign language gestures were gathered from a variety of sources to create a broad dataset. The collection contains a sizable number of photos with signs in sign language. The sign language word that each gesture corresponds to has been labeled. The dataset underwent preprocessing, which involved scaling all photos to a common resolution of pixels, before training. Additionally, to guarantee consistent input for the model, pixel values were standardized to a range of [0, 1]. To increase the dataset's variety and strengthen the model's robustness, data augmentation techniques including random rotations and horizontal flips were used.

The convolutional neural network (CNN) architecture called Xception was selected as the model architecture for this study due to its intricate and effective design. Since good gesture detection depends on the capacity to record complex picture information, Xception was chosen. The model was able to take the benefit of previously learned features since the weights were initialized using pre-trained weights on a sizable picture classification dataset. To adapt the pre-trained Xception model for the specific task of gesture recognition, the original classification layer was replaced with a new layer that corresponds to the number of distinct sign language gestures. Except for the recently added classification layer, all the Xception model's layers were frozen during the initial training phase.

This method allowed the model to specialize for the present job while preserving the useful features discovered during pre-training. Training (80%), validation (10%), and test (10%) subsets were each given 10% of the total dataset. The difference between expected and real gestures was measured using the categorical cross-entropy loss function. The model's weights were modified throughout training using the Adam optimizer with a learning rate of 0.001. On the validation set, the model's performance was observed, and early stopping was used to avoid overfitting.

The Xception model was fine-tuned by gradually unfreezing some of its layers to improve the performance of the model. On the test set, the trained model was evaluated to see how well it could predict sign language movements in real-world scenarios. To measure the performance of the model, accuracy metrics were produced.

B. Amplifying Potential Empowering Hearing-Impaired Children through Sound and Transcription

Sound Amplification is a feature of the Hearing-Impaired Application that allows the child to customize a sound amplifier to suit their own unique hearing needs. It adapts to your hearing level to provide optimal amplification while reducing background noise. Additionally, the user can obtain transcriptions of any audio file uploaded to or recorded on their device. By providing a variety of customization options, Sound Amplification aims to enhance the user experience and promote inclusion for those with hearing loss. The constructed transcription model utilizes unsupervised learning

The goal of sound amplification is to make audio in online apps clearly understood and more personalized. Without the need for other frameworks, it does this utilizing JavaScript and the Web Audio API. The idea of sound amplification, Helps the user to easily understand any recorded soundtrack which allows the user to customize the loudness and speediness as their own, it is the main point of interest. The article uses the.media Devices.get User Media API, which enables access to the user's microphone input for audio recording, to record these recordings. The paper shows how the GainNode of the Web Audio API is crucial in increasing the loudness once audio is collected. By acting as an amplifier, this GainNode enables users to modify the power of the audio signal. This strategy's accessibility and simplicity are what make it so beautiful. Developers obtain a better grasp of audio processing by depending only on the Web Audio API and JavaScript. The lack of external frameworks guarantees a clearer understanding of the core ideas, making it both an instructive and useful endeavor.

This research gives developers the tools they need to give users more individualized and impactful audio experiences. This research reveals the ways to successfully enhance audio messaging, audio playing, or any application demanding increased aural impact. In the end, the research enhances web-based interactions by utilizing more colorful and captivating acoustic content.

This study involves the utilization of the Transcript function for the purpose of audio analysis. Specifically, the application is employed to remove non-speech components from an audio file and generate an audio list. This audio list is subsequently embedded through the use of the Voice Encoder function, which is a feature of the resemblyzer python package. In this study, audio data were subjected to clustering based on their embedding using Spectral Clustering. The resulting clusters were then identified as corresponding to individual speakers. The subsequent step in the process involves the conversion of each audio component utilizing a speech-to-text model, resulting in the extraction of the corresponding textual content. The present study concerns the speech-to-text model, which is a type of sequence-to-sequence transformer model. This research concerns the inputs and outputs of a speech-to-text system. Specifically, the inputs consist of audio segments, while the outputs consist of transcripts of the previously mentioned audio segments.

C. Sound Classification to identify the surrounding while intercepting learning to Hearing impaired children

Sound classification is of utmost significance in the IHear mobile application due to its ability to provide hearing-impaired children with an enriched and immersive learning experience. It enables IHear to recognize and classify a variety of environmental sounds. This feature allows children to become more attuned to their auditory environment, resulting in a more profound understanding of the world. Children can perceive context and respond appropriately if they can distinguish and comprehend various sounds. Accurate speech perception is crucial for language development. The sound classification component facilitates sound perception ability development by allowing children to recognize and distinguish speech sounds from other background noises. This

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facilitates enhanced speech articulation and language comprehension.

The IHear application's Sound Classification feature employs the ESC-50 dataset, a collection of 50 distinct classes of environmental sounds, to forecast and amplify auditory perception. For audio classification, Google developed YAMNet, a pre-trained convolutional neural network (CNN) that is easily accessible on TensorFlow Hub. The YAMNet model has been instantiated, and the necessary libraries, namely TensorFlow, TensorFlow IO, and TensorFlow Hub, have been imported. In order to adhere to the IEEE format, this research defines a utility function that serves the purpose of loading and resampling WAV files. This functionality involves the acquisition of the ESC-50 dataset, which has been obtained and subsequently processed by reading its metadata in CSV format to prepare the dataset paths.

After the creation of a TensorFlow dataset object using the filenames, targets and folds, A function is utilized to convert the audio files in the dataset to a YAMNet-compatible format. The YAMNet model extracts the embeddings, which can also be considered low-dimensional feature vectors and stores them as a new dataset.

The dataset with embeddings is taken as input for the developed model. The model is a Multilayer Perceptron (MLP), which is a straightforward neural network. The Multilayer Perceptron (MLP) model is an architecture of neural networks consisting of one or more hidden layers and an output layer. The hidden layers are responsible for transforming the input data into a format that can be utilized by the output layer. The output layer, on the other hand, generates the model's ultimate output. The output layer has an equivalent number of units to the total number of distinct classes. The model generates predictions for the class label of each audio file. The training dataset is used to train the model, while the validation dataset is used for model validation. The training methodology incorporates early stopping, a technique that monitors validation loss to prevent overfitting. In the process of model evaluation, the independent test dataset is utilized to test the model's performance, and subsequently, the loss and accuracy metrics are computed and presented.

This integration enables the combination of YAMNet's audio classification proficiency with the adaptability and knowledge acquisition potential of an MLP. This has a positive impact on the customized learning experience of children who have hearing disabilities.

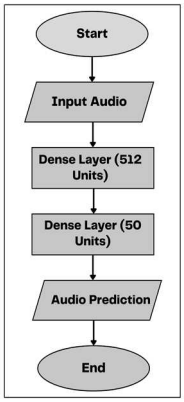


Fig. 2. Flowchart of Sound Classification

D. Interactive and educational activity to enhance the hearing through sound familiarities

In this Speaking accuracy activity involves several steps to assess the similarity of pronunciation between user audio files and a base case. First, a filter is applied to the user audio to remove background noises [13]. This is achieved by utilizing two Butterworth filters: a lowpass filter and a highpass filter. These filters help eliminate unwanted frequencies and ensure that the focus remains on the relevant speech signals.

Next, spectrograms are created for both the user audio and the base case audio files [14]. The audio data is divided into segments of 256 length, and spectrograms are generated for each segment. Spectrograms provide a visual representation of the frequency content over time in the audio signal. By analyzing the spectrograms, it becomes possible to identify the specific characteristics and patterns of speech pronunciation

To measure the similarity between the two spectrograms, pairwise Canberra distance is calculated [15]. The Canberra distance is a measure of dissimilarity between two sets of data, and in this case, it is used to quantify the difference in pronunciation between the user and the base case. By computing the Canberra distance for each pair of spectrograms, a distance matrix is constructed [16].

To normalize the distance matrix and ensure consistent scaling, normalization techniques are applied. This normalized distance matrix serves as the basis for determining the score of the similarity of pronunciation between the two audio files. The normalized scores provide a quantitative measure of how closely the user's pronunciation aligns with the base case.

A threshold is set to determine whether the user's pronunciation is considered close enough to the base case. If the score falls below the threshold, it indicates a sufficient level of similarity in pronunciation. This threshold serves as a criterion for evaluating the accuracy of the user's pronunciation.

The research utilizes several software packages to facilitate the implementation of the methodology. Scikit-Learn, a machine learning library [17], is utilized for various tasks, including distance calculations. Scipy, a scientific computing library, provides the necessary tools for signal processing, including the application of Butterworth filters [18]. Additionally, Matplotlib, a plotting library, is employed for visualizing the spectrograms.

By following this methodology, the research aims to objectively assess and quantify the similarity of pronunciation between user audio files and a base case. The findings will provide valuable insights into the effectiveness of the background noise filtering techniques, distance measurement methods, and threshold determination in evaluating the accuracy of pronunciation.

IV. Results and Discussion

In this research, we conducted a comprehensive evaluation of the IHear Application's performance, particularly focusing on the models designed for sound classification, hearing screening, speech accuracy, and sound amplification. The performance of the trained models on a test dataset was evaluated to obtain classification accuracy from the gathered data.

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The application comprises both supervised and unsupervised models. Supervised models are utilized when predefined classes or outcomes are known, making them ideal for tasks that require a precise understanding of class boundaries. The Sound Classification Model and the Sign Language Model under screening functionality operates under the umbrella of supervised learning.

The Sound Classification Model within the IHear mobile application plays a pivotal role in accurately categorizing and understanding a wide range of auditory stimuli. The evaluation of this model revealed an impressive accuracy rate of 61.10%, signifying the model's ability to precisely classify various sounds in its auditory environment. To provide a visual representation of this accomplishment, Fig. 3 and Fig. 4 are utilized to depict the sound classification model. These figures analyze the impact of the number of training iterations, referred to as 'Epochs,' on the model's accuracy. Fig. 3 and Fig. 4 provide a visual representation of the model's performance across four key parameters: training accuracy, validation accuracy, training loss, and validation loss.

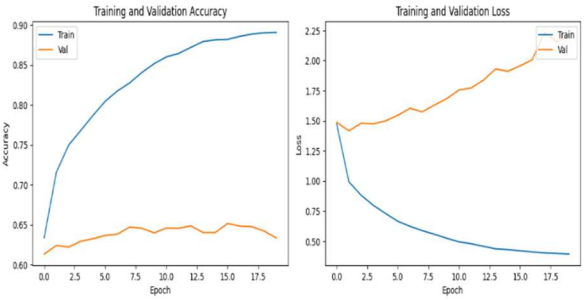


Fig. 3. Sound Classification Training and Validation Accuracy and Loss

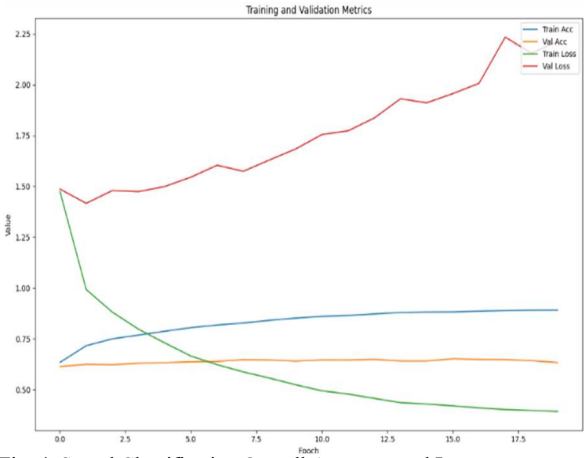


Fig. 4. Sound Classification Overall Accuracy and Loss

In the context of the IHear mobile application, which aims to facilitate a positive and constructive interaction with auditory stimuli, a model was utilized to evaluate the communication abilities of children with mild to moderate hearing impairments using sign language.The model serves as a benchmark for a child's ability to communicate with the external environment, considering that the targeted users have a low level of speech perception. The developed model was determined to be of 92 percent.

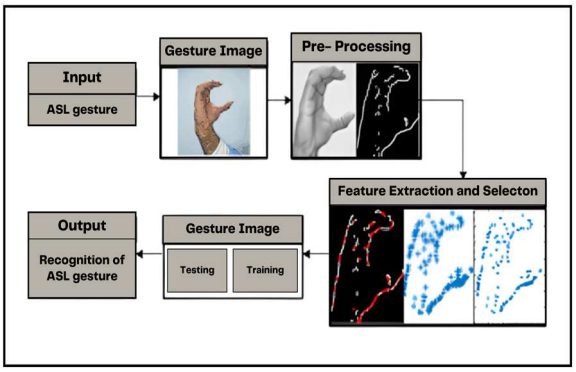


Fig. 5. Summarized Screening Function Results

In Fig. 6, the accuracy, precision, recall, and loss of the hand sign screening model are depicted in relation to the number of epochs.

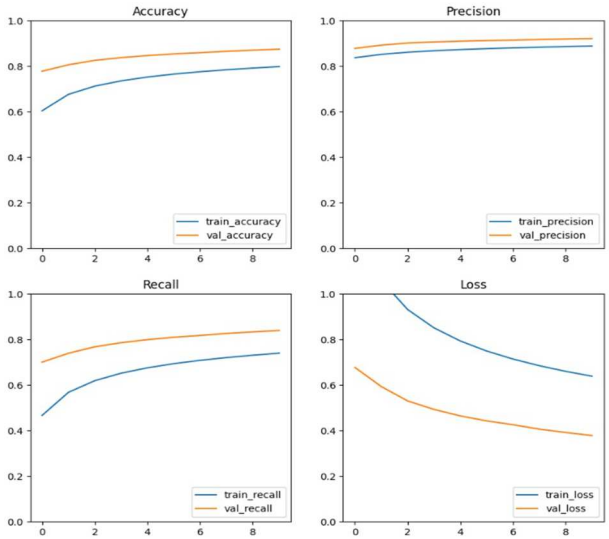


Fig. 6. Accuracy, Precision, Recall and Loss of Screening Model

The unsupervised models are designed to identify patterns and relationships within the data without the need for labeled data. These two unsupervised models encompass the amplification function and speech accuracy checking function. .Overall, the results provide insightful information regarding the efficacy of the trained models.

V. Conclusion

IHear is an innovative application designed to meet the specific educational requirements of hearing-impaired children by providing a variety of features and functions. IHear is an outstanding representation of the significant advancement in the field of assistive technology and has had an immense effect on the lives of these children. By providing them with the necessary tools and resources, the provision of a personalized learning experience to enhance children's communication skills, using cutting-edge technologies has the potential to significantly improve the quality of life and academic performance of children with speech and hearing impairments. Additional research and development in these areas will contribute to the enhancement of language learning technology and its effectiveness for language learners. This study demonstrates the potential for IHear to advance and

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leave an enduring mark in the field of education for children with speech and hearing impairments. Through ongoing research and development, the application can continue to advance and increase its effectiveness in meeting the requirements of this population.

VI. Future Work

The IHear application is a promising educational solution for children with speech and hearing impairments. The present study has yielded significant insights regarding the application's efficacy and potential impact. It highlights several prospective research avenues that could enhance the application's capabilities and benefits. There is a pressing need to investigate the expansion of IHear's language support to accommodate a linguistically diverse user base. This necessitates researching the most effective methods for incorporating a wider variety of languages, thereby enhancing usability and accessibility. This enhancement could boost the application's overall efficacy and expand its reach.

Despite the fact that IHear presently focuses predominantly on hearing impairments, we hope to incorporate sign languages into its educational repertoire in the future. This expansion aims to provide a setting that comprehensively meets the communication requirements of children with various linguistic and aural challenges. As the first step in exploring the sign language avenue, IHear has touched on the subject under the screening component. By establishing a balance between accessibility and sustainability, we hope to increase IHear's availability, thereby empowering more hearing-impaired children with valuable educational resources.

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